YIXIAN ZHANG

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EDUCATION

Tsinghua University *Aug 2024 - Present*
M.S. in Information Science and Technology GPA: 3.96/4.0

Southeast University *Sep 2020 - June 2024*
B.S. in Information and Computing Science GPA: 3.8/4.0
Relevant Coursework: Reinforcement Learning, Robotics, AI Agents, Convex Optimization

SELECTED PUBLICATIONS

Research Area: Sample-Efficient Reinforcement Learning

- **SAC Flow: Sample-Efficient Reinforcement Learning of Flow-Based Policies** **ICLR 2026**
  Yixian Zhang *(First Author), S. Yu, C. Yu, W. Ding*
  Treats flow policy as a sequential model; utilizes GRU/Transformer velocity reparameterization to stabilize off-policy training and boost sample efficiency.
  Project Page GitHub Code
- **Policy Newton Algorithm in Reproducing Kernel Hilbert Space** **ICLR 2026**
  Yixian Zhang *(First Author), H. Tang, C. Wei, C. Wang, W. Ding*
  Proposes a non-parametric, second-order RL scheme that significantly enhances sample efficiency and optimization speed of policy gradient methods.
- **Residual Kernel Policy Network: Enhancing Stability in RKHS-Based RL** **ICLR 2025**
  Yixian Zhang *(First Author), H. Tang, H. Lin, W. Ding*
  Introduces an RKHS policy with residual design and representation learning, achieving improved stability and data efficiency in continuous control tasks.
- **Distributional Decision Transformer: Risk-Sensitive Offline RL** **IROS 2025**
  C. Wei, H. Tang, ***Yixian Zhang****, C. Wang, W. Ding*
  Developed quantile-based critics for risk-sensitive offline reinforcement learning.

Research Area: Large Multi-Agent & Networked Systems

- **High-Dimensional Probability Preserving Scenario Reduction** **Applied Energy**
  Yixian Zhang *(First Author), Y. Tang, J. Hu, et al.*
  Developed a scenario reduction method preserving high-dimensional probability structures for scalable multi-agent power networks.
- **Deep Reinforcement Learning using Optimized Monte Carlo Tree Search** **IEEE ToG**
  Yixian Zhang *(First Author), Z. Li, Y. Cao, X. Zhao, J. Cao*
  Combined DRL with optimized MCTS to solve competitive multi-agent gameplay challenges in EWN.

OPEN SOURCE PROJECTS

- **RLinf: RL for Embodied and Agentic AI** RLinf/RLinf
  Open-Source Contributor arXiv 2509.15965 **2.6k**
  A flexible and scalable RL infrastructure supporting post-training of foundation models (LLMs, VLMs, VLAs) for embodied and agentic AI. **Solely designed and implemented** the full DSRL (π_0) training pipeline within RLinf, enabling efficient end-to-end RL fine-tuning of the π_0 VLA model. **Co-contributed** to the $\pi_{0.6}$ training integration and the SAC Flow component, extending RLinf's support for flow-based policy optimization in embodied settings.

HONORS & AWARDS

- **Outstanding Graduate of Southeast University** (Top 5%) *June 2024*
- **Excellent Graduation Thesis**, Southeast University (Top 5%) *June 2024*
- **First Prize**, China Undergraduate Mathematical Contest in Modeling (Top 0.5%) *2022*